Nallain Zahra

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Google Scholar: *[View Profile]*

Academically driven graduate in Chemistry and Conservation Management with teaching, research, and laboratory experience. Seeking a role that allows me to contribute to research and student learning in an academic environment.

Education

2023-2024 **MS in Conservation Management**
Edgehill University, United Kingdom

2019–2021 **MPhil in Chemistry**
University of Gujrat, Pakistan

2014–2018 **BS (Hons) in Chemistry**
Chemistry GC University Faisalabad, Pakistan

Professional Teaching Qualification

2018-2020 **Bachelor of Education (B.Ed., 1.5 Years)**, *Major in Science Education*
Allama Iqbal Open University, Islamabad, Pakistan

Research Work

2020- 2021 Comparison between Nuclease Activity of Sulfonamides and Their Metal Complexes

2023-2024 Exploring Biodiversity in Urban Parks: A Focus on Avian Species across Different Habitats

Work Experience

- Teacher at
  - Sialkot Grammar High School, Pakistan
  - Allied School Sialkot, Pakistan
  - Cadet College Sialkot, Pakistan
  - Scholars Academy Sialkot, Pakistan
- Team member in Taco Bell-East Didsbury, Manchester
- Free Lancing in Environmental Sciences Project
- Online Tutor to various International Students

Professional Skill

- Continuous Professional Development
- Health and Safety Standards

- Teamwork
- Staff Education and Training
- Product Knowledge

Experimental Techniques

- **Gel Electrophoresis:**

  Proficient in gel electrophoresis techniques for separating DNA .This includes experience in preparing gels, loading samples, running electrophoresis, and visualizing separated bands.

- **Synthesis of Compounds:**

  Possess hands-on experience in the synthesis of metal sulfonamides. This experience encompasses following reaction protocols, handling chemicals safely, performing purification techniques, and isolating desired products.

- **Physical Characterization:**

  Skilled in determining the melting point of synthesized organic compounds. This includes using appropriate equipment and interpreting the results to assess compound purity.

Publications

- Danish, M., Bibi, A., Akhtar, A., Noreen, N., Batool, F., **Zahra, N**. & Asiri, A. M. (2022). Theoretical and Experimental Investigations of N-and O-Alkylated Sulfonamides: Density Functional Theory, Hirshfeld Surface Analysis, and Molecular Docking Studies. Chemistry Select, 7(4), e202103209.
- Ishtiaq, R., **Zahra, N.**, Iftikhar, S., Rubab, F., Sultan, K., Abbas, A., . & Park, K. Y. (2024). Adsorption of Cr (VI) ions onto fluorine-free niobium carbide (MXene) and machine learning prediction with high precision. Journal of Environmental Chemical Engineering, 12(2), 112238.
- Iftikhar, S., **Zahra, N.,** Rubab, F., Sumra, R. A., Khan, M. B., Abbas, A., & Jaffari, Z. H. (2023). Artificial neural networks for insights into adsorption capacity of industrial dyes using carbonbased materials. Separation and Purification Technology, 326, 124891.
- Jaffari, Z. H., Iftikhar, S., Ishtiaq, R., **Zahra, N.**, Rubab, F., Lam, S.M., Abbas, A., Probabilistic prediction of Adsorption Capacity for phosphate ion removal onto biochar materials using machine learning algorithms with a large dataset. Journal of Hazardous Materials. Under revision.
- Abbas, G. Rubab, F., **Zahra, N.,** Abbas, A.,Peroxymonosulfate axtivation of novel catalyst $Cd_{0.4}Cu_{0.3}Mn_{0.3}S@Ti_3AlC_2$ 2D-Max-phase for diclofenac sodium degradation: DFT and machine learning analysis. Applied Catalysis. Under revision.

Other Skills

- Data collection and Analysis
- Research Management
- Laboratory and Analytical Techniques

Language Test

IELTS- Band Score-6.5